

Investigating radiation heat transfer from the cone calorimeter heater: A new view factor model and uncertainty quantification

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Abstract

This work seeks to better characterize the radiant heat transfer to a sample in the cone calorimeter, a widely used fire testing apparatus. Specifically, the spatial uniformity of the heat flux from the cone heater to a sample and to a heat flux sensor are investigated with analytical view factor models based on the idealized geometry of cone heater element (tapered helical coil) and experimental measurements. The view factors are calculated using the contours of the relevant geometries and applying Stokes' theorem, with the contour integrals evaluated numerically. An uncertainty analysis is performed on the theoretical incident heat flux to evaluate the reliability of the model by comparing predicted and experimental values. The incident heat flux to the sample surface is measured using a water-cooled radiometer, while the temperature spatial variation of the cone heater surface is determined through color-ratio pyrometry thermograms with a digital camera. The measurements are used to showcase the proposed formulation. The findings contribute to a better understanding of the cone calorimeter heater view factor model, offering valuable insights for researchers and engineers seeking improved accuracy in fire safety assessments.

Keywords: Cone heater, view factor, radiant heat fluxes, ratio pyrometry

1. Introduction

The cone calorimeter is a widely used instrument for fire testing and research, developed at the National Bureau of Standards (NBS) [1], the predecessor to the National Institute of Standards and Technology (NIST). It is designed to measure key flammability parameters under controlled conditions, including heat release rates, ignition time, emissions, and mass loss rate [2]. These measurements provide insights into the fire behavior of materials, fostering the development of safer, more fire-resistant products.

A key component of the cone calorimeter is its radiant heater in the shape of a tapered helical-coil. The heater subjects the sample to a thermal insult similar to that expected in real fires ($\sim 100 \text{ kW}\cdot\text{m}^{-2}$) [2]. The heater is designed to minimize convective heating, ensuring uniform irradiance across the sample surface. Both the ISO 5660 and ASTM E1354 standards specify that the irradiance across the central $50 \times 50 \text{ mm}^2$ area of the specimen's exposed surface must not deviate by more than $\pm 2\%$ under the standard configuration [2–4].

Ensuring spatially uniform irradiance from the radiant heater is essential for achieving a quasi-1D heating configuration in fuel samples, as required by the ISO 5660 [3] and ASTM E1354 [4] standards. Studies have focused on characterizing the heater's thermal performance and developing models to predict the heat flux distribution. Boulet et al. [5] measured radiation intensity and spectral emission from various regions of the cone heater to assess non-uniformity in its surface temperature. They observed temperature discrepancies of approximately 200 K

across the cone surface when the setpoint was $50 \text{ kW}\cdot\text{m}^{-2}$, highlighting significant heterogeneities. These findings demonstrated that the cone's surface does not emit uniformly, leading to variations in the radiant heat flux received by the sample.

In previous studies, a truncated cone (TC) configuration has been assumed as an "idealized" geometry for representing the cone calorimeter radiant heater [6–8]. However, the TC approach for estimating the view factor does not consider the geometrical complexities of the cone heater, resulting in an underestimated non-uniformity in the incident heat flux to the sample.

This work proposes new view factor formulations from the sample surface and a sensor to the cone heater. The formulation uses an idealized geometry: a tapered helical coil (HC) for the cone calorimeter's heater to calculate the view factor between a target and the heater. The HC model is benchmarked against the TC model across various scenarios. Color ratio pyrometry is used to determine the temperature spatial variation from thermograms in the cone heater, while theoretical incident heat fluxes are validated against experimental radiometer measurements. An uncertainty analysis is performed for the view factors, temperature spatial variation, and experimental measurements to better understand the variations and limitations associated with both the HC and TC approaches.

2. Methodology

2.1. Radiative heat fluxes

The incident radiation to the target, represented by the heat flux gauge sensor area and denoted as $\dot{q}_{inc,g}''$, is related to the power

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Nomenclature

Latin symbols

$\dot{q}_{inc}''$	Incident radiative heat flux ($\text{kW}\cdot\text{m}^{-2}$)
$\dot{q}_{emi}$	Radiation emitted (W)
A	Surface area (mm^2)
d	Vertical distance (mm)
D_{elem}	Heater element diameter (mm)
F	View factor (-)
H	Cone heater height (mm)
I	Interval
L	Sample length side (mm)
P	Percentage variation (%)
p	Sensor diameter and small element diameter (mm)
R_k	Radial distance (mm)
R_{ij}	Euclidean distance between the contours i and j (mm)
S	Calibration constant of the radiometer ($\text{V}\cdot(\text{kW}/\text{m}^2)^{-1}$)
T	Temperature ($^{\circ}\text{C}$ or K)
V	Voltage (V)

Greek symbols

λ	Wavelength (nm)
α	Angle (rad)

δ	Uncertainty (a.u.)
Γ	Contour
σ	Stephan-Boltzmann constant ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$)
σ_i	Standard deviation of i (a.u.)
ε	Emissivity (-)

Acronyms

CRP	Color ratio pyrometry
FOV	Field of view
G/R	Green-to-red intensity ratio
HC	Helical coil
OID	Object-to-image distance
RMSE	Root mean square error
TC	Truncated cone

Subscripts

bb	Blackbody radiation source
c	Cone calorimeter heater
g	Heat flux gauge sensor
s	Sample

emitted from the cone heater, $\dot{q}_{emi,c}$, by the following expression:

$$\dot{q}_{emi,c} F_{c \rightarrow g} = A_g \dot{q}_{inc,g}'' \quad (1)$$

Here, $F_{c \rightarrow g}$ is the view factor from the cone heater to the gauge's sensor. By applying the reciprocity rule of view factors, $A_c F_{c \rightarrow g} = A_g F_{g \rightarrow c}$, Eq. (1) can be rearranged to express $\dot{q}_{inc,g}''$:

$$\dot{q}_{inc,g}'' = F_{g \rightarrow c} \dot{q}_{emi,c}'' = F_{g \rightarrow c} \varepsilon_c \sigma T_c^4 \quad (2)$$

In this equation, $F_{g \rightarrow c}$ is the view factor from the gauge's sensor (g) to the cone heater (c), while ε_c denotes the emissivity of the cone heater surface, σ is the Stefan-Boltzmann constant, and T_c is the temperature of the cone heater. For this study, the cone heater emissivity is assumed to be 0.99 [9].

The characterization of incident radiant heat fluxes will be conducted through experimental measurements, while an analytical expression for $F_{g \rightarrow c}$ will be derived based on an idealized helical coil (HC) representation of the cone heater geometry. This derived expression will then be compared to the truncated cone (TC), which has been assumed as the idealized geometry for representing the cone calorimeter heater in previous studies [6–8].

2.2. Mathematical formulation of View Factor Model

The view factor between two finite surfaces, A_i and A_j , can be calculated mathematically from their contours Γ_i and Γ_j , respectively, by applying Stokes' theorem twice to the view factor

definition (given by the surface integrals) [10, 11]:

$$F_{i \rightarrow j} = \frac{1}{2\pi A_i} \oint_{\Gamma_j} \oint_{\Gamma_i} \ln(R_{ij}) dr_i dr_j \quad (3)$$

Here, R_{ij} represents the Euclidean distance between the contours defined by the positions $\mathbf{r}_i$ and $\mathbf{r}_j$, respectively. This contour methodology is utilized to determine the view factor between the cone heater and targets (e.g., radiometer or small patch of a sample surface) positioned at various locations from the cone heater.

Figure 1 shows an isometric view of the two idealized cone heater geometries: the helical coil (HC; left), and the truncated cone (TC; right). To improve visualization of the internal structure, transparency is applied for all regions of the idealized heater geometries where $y < 0$. According to the ISO 5660 standard [3], $H = 65$ mm represents the vertical height of the cone heater, $R_1 = 40$ mm defines the radial distance at the top of the cone heater (at $z = H$), and $R_1 + R_2$ defines the radial distance at the lower (base) section of the cone heater, with $R_2 = R_1$.

In the HC configuration, the heater element is coiled in along a helical path. This results in a vertical displacement of $H/19$ for the bottom coil along the x -axis after it has undergone a half turn. This feature is a results from the coil's geometry, but is absent in the TC configuration. Both configurations consider the thickness of the cone heater base plate, $d_2 = 5$ mm. However, the base plate surface was not considered as a participating surface in the radiation analysis. Below this plate, a target (i.e., a

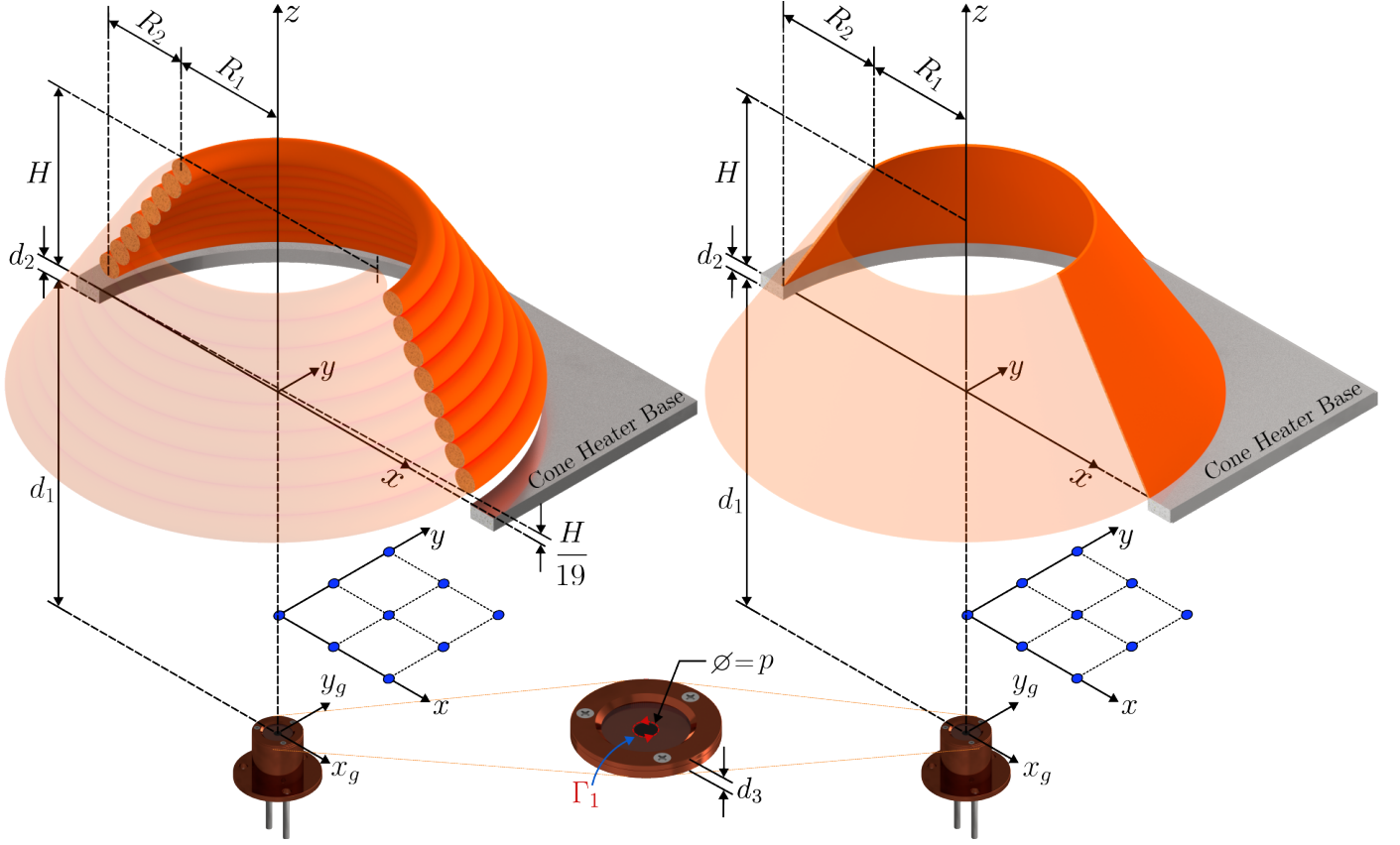


Figure 1: Schematic of the experimental setup and dimensions for the cone calorimeter heater (not to scale), illustrating two idealized geometries: a helical coil arrangement (left) and a truncated cone (right). The constants correspond to $R_1 = R_2 = 40$ mm and $H = 65$ mm, which are taken from the ISO 5660 standard [3].

sample or heat flux gauge) is placed at a distance d_1 . A heat flux gauge (radiometer) is shown in the figure with the vertical distance from the top of the gauge to the sensor marked as d_3 , indicating the sensor's depth from the sapphire window and copper body. This value is specific to the gauge and the value in this work was 3 mm. To characterize the incident heat flux, the heat flux gauge was moved within the (x, y, z) coordinate system. The blue markers in the figure indicate measurement points where the gauge was positioned in the first quadrant of the (x^+, y^+) plane, representing a subset of the positions used in the experiments. In total, 25 measurement points were used across all 4 quadrants.

2.2.1. View factor from the cone heater to a disk

The mathematical formulation of the view factor in this work requires a parametric representation of the surfaces of the cone heater and the target, constructed from their respective contours. Consistent with the sensor design, the gauge's sensor area is assumed as a disk (see Fig. 1) [12, 13]:

$$\Gamma_1 : \mathbf{r}_1(u) = \left(\frac{p}{2} \cos u + x_g, \frac{p}{2} \sin u + y_g, -d \right), \quad 0 \leq u \leq 2\pi \quad (4)$$

Here, p is the diameter of the disk, x_g and y_g denote the specific locations within the $(x, y, -d)$ domain where the heat flux gauge is placed (cf. Fig. 1). The term d corresponds to the vertical distance along the z axis, where $d = d_1 + d_2 + d_3$. It

is worth noting that when the target is the sample surface, d_3 is considered to be zero. However, for the radiometer used in this study, $d_3 = 3$ mm. To model the incident heat flux to a target from the cone heater, its geometry must first be understood. The cone heater is constructed from a long cylindrical element wound into a tapered helical coil [3, 4], as shown in Fig. 2a. Previous studies have approximated this geometry as a truncated cone [6–8]. In this work, the cone heater's contour is modeled piecewise using two tapered helices and two semi-circumferences as the piecewise contours. Specifically, each helix is represented parametrically as follows:

$$\mathbf{r}(v) = (f(v) \cos v, f(v) \sin v, bv + c), \quad v \in I, \quad (5)$$

where $f(v) = r_0 - av$ is a linear function. The parameter r_0 is the initial maximum radial distance of the coil, a controls the linear radial decrease, b sets the vertical pitch along the z -axis, and c specifies the initial vertical offset. Together, these parameters shape the radial profile and vertical extension of the helix. The following equations determine these parameters (a, b, c and r_0) from the initial and terminal coordinates of the cone heater.

$$\mathbf{r}(0) = (-R_1 - R_2, 0, 0), \quad \mathbf{r}(19\pi) = (R_1, 0, H)$$

Given these conditions, the parameters (a, b, c and r_0) can be determined by solving a system of linear equations. The cone

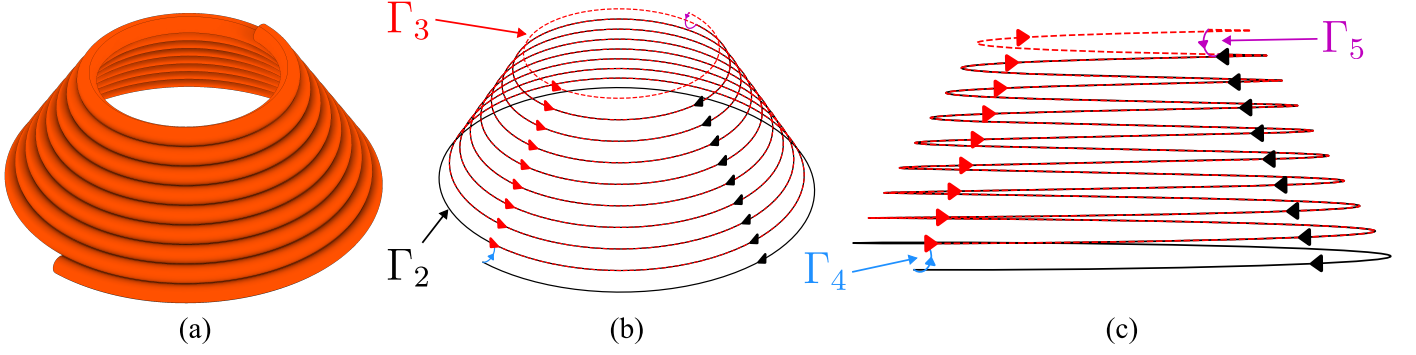


Figure 2: (a) Isometric 3D model showing idealized helical coil geometry, (b) Isometric view of Γ_2 and Γ_3 contours of helical coil and curve (cf. Eq. 6) orientations (represented by arrows), and (c) Side view of (b) showing Γ_4 and Γ_5 (cf. Eq. 9), which are the semi-circumferences.

heater is defined by the curve from Eq. (5), which is duplicated across different domains to ensure closure with two semi-circumferences at the beginning and end of its geometry, as shown in Fig. 2c.

$$\begin{aligned}\Gamma_2 : \mathbf{r}_2(v) &= \left(f(v) \cos v, f(v) \sin v, \frac{Hv}{19\pi} \right), & 17\pi \leq v \leq 0 \\ \Gamma_3 : \mathbf{r}_3(v) &= \mathbf{r}_2(v), & 2\pi \leq v \leq 19\pi\end{aligned}\quad (6)$$

Between the two helices (Γ_2 and Γ_3), there is an offset in v of 2π radians and each coil completes 8.5 turns. The linear function, $f(v)$, is given by:

$$f(v) = -R_1 - R_2 + \frac{R_2 v}{19\pi} \quad (7)$$

The diameters of the semi-circumferences (Γ_4 and Γ_5 in Fig. 2c) are the diameter of the heater element, D_{elem} , which is related to other features of the geometry:

$$D_{elem} = \|\mathbf{r}_2(2\pi) - \mathbf{r}_2(0)\| = \frac{2\sqrt{R_2^2 + H^2}}{19} \approx 8 \text{ mm} \quad (8)$$

This result was validated through caliper measurements taken at multiple locations across several cone heaters. The semi-circumferences to close the contour of the helical surface are described by:

$$\begin{aligned}\Gamma_4 : \mathbf{r}_4(v) &= \left(\frac{D_{elem}}{2} \cos v - R_1 - \frac{18R_2}{19}, \frac{D_{elem}}{2} \sin v + \frac{H}{19} \right) \\ \Gamma_5 : \mathbf{r}_5(v) &= \left(\frac{D_{elem}}{2} \cos v + R_1 + \frac{R_2}{19}, \frac{D_{elem}}{2} \sin v + \frac{18H}{19} \right)\end{aligned}\quad (9)$$

The domains of the parametric representations of Γ_4 and Γ_5 are $\alpha - \pi \leq v \leq \alpha$ and $\pi - \alpha \leq v \leq 2\pi - \alpha$, respectively, where $\alpha = \arctan(H/R_2)$. The distance between the contours (R_{1j}) and the dot product between the tangent vectors ($d\mathbf{r}_1 d\mathbf{r}_j$) are given by:

$$\begin{aligned}R_{1j} &= \|\mathbf{r}_1 - \mathbf{r}_j\|, \quad j = 2, 3, 4, 5. \\ d\mathbf{r}_1 d\mathbf{r}_2 &= \frac{p}{2} (f(v) \cos(v-u) + f'(v) \sin(v-u)) du dv \\ d\mathbf{r}_1 d\mathbf{r}_3 &= d\mathbf{r}_1 d\mathbf{r}_2 \\ d\mathbf{r}_1 d\mathbf{r}_4 &= d\mathbf{r}_1 d\mathbf{r}_5 = \frac{p\sqrt{R_2^2 + H^2}}{38} \sin u \sin v du dv,\end{aligned}\quad (10)$$

Finally, the view factor from the sensor to the cone heater is given by:

$$\begin{aligned}F_{g \rightarrow c} &= \frac{2}{\pi^2 p^2} \left(\int_{17\pi}^0 \oint_0^{2\pi} \ln(R_{12}) d\mathbf{r}_1 d\mathbf{r}_2 + \int_{2\pi}^{19\pi} \oint_0^{2\pi} \ln(R_{13}) d\mathbf{r}_1 d\mathbf{r}_3 \right. \\ &\quad \left. + \int_{\alpha-\pi}^{\alpha} \oint_0^{2\pi} \ln(R_{14}) d\mathbf{r}_1 d\mathbf{r}_4 + \int_{\pi-\alpha}^{2\pi-\alpha} \oint_0^{2\pi} \ln(R_{15}) d\mathbf{r}_1 d\mathbf{r}_5 \right)\end{aligned}\quad (11)$$

This expression is numerically solved with negligible integration error, on the order of $O(10^{-9})$, using adaptive methods in both Mathematica and Python, with Python employing Gaussian quadrature.

2.3. Incident radiative heat flux measurements

To isolate the radiative component of the total incident heat flux, sapphire windows are placed atop heat flux gauges. Sapphire's transparency to infrared radiation in the 0.125–5.5 μm spectral region [14] makes it ideal for measuring radiative heat flux without significant absorption or distortion. These windows effectively block convective heat transfer, ensuring the accurate capture of radiative incident heat flux [15].

A Medtherm 64P-15-24 Gardon Gauge with a sapphire window (referred to as the radiometer) was integrated into a traverse system within the Firetesting Technology iCone Classic apparatus to map horizontal planes at varying vertical heights (d_1 in Fig. 1). The system was controlled by an Arduino UNO R3 with a stepper-motor shield, interfaced with LabVIEW for automation. Data were collected at 25 points per plane (spaced 25 mm apart) across a $100 \times 100 \text{ mm}^2$ area, corresponding to ASTM E1354 and ISO 5660 standard sample sizes [3, 4]. The radiometer moved along the x and y axes, and 70 seconds of voltage data were recorded at 4 Hz at each point. After each set of measurements, the radiometer was repositioned vertically to the next height. Measurements were performed for three cone heater temperatures ($T_{c,PID} \in 750, 800, 850^\circ\text{C}$), which represent the set temperatures at the PID controller of the cone calorimeter, at two heights ($d_1 = 25$ and $d_1 = 60 \text{ mm}$) and at eight different heights along the cone heater centerline $(0, 0, z)$. The first 10 seconds of data were excluded to account for potential vibrations, leaving 60 seconds of data for analysis.

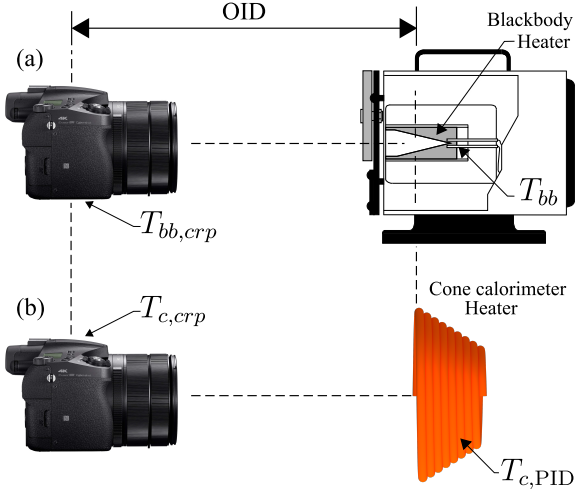


Figure 3: Object-to-image distance (OID) is consistent for both configurations: a) camera positioned relative to the blackbody heater, and b) camera aimed at the vertically oriented cone heater (not to scale). In this context, $T_{c,PID}$ represents the cone heater temperature set at the PID controller, while T_{bb} denotes the temperature of the blackbody heater. The temperatures obtained using the two color-ratio pyrometry technique are represented by $T_{bb,crp}$ for the blackbody and $T_{c,crp}$ for the cone.

2.4. Color-ratio pyrometry

Color-ratio pyrometry (CRP) offers a cost-effective, non-intrusive optical technique for temperature measurement. CRP analyzes the spectral intensity of radiation emitted by a hot object and, based on the Planck distribution, measures the ratio of radiation intensities at two distinct wavelengths. This reduces the influence of emissivity and allows for precise temperature measurements [16–18]. CRP has applications in various fields, including the measurement of soot temperatures in flames [19–21] and glowing embers and sparks [22–25], among others. This technique is employed to measure the temperature of the cone heater and its spatial variation.

2.4.1. Camera calibration and measurements

A Sony DSC-RX10 IV DSLR camera was used for color-ratio pyrometry. Calibration was performed with a blackbody source heater (Infrared Systems Development IR-564/301) at temperatures ranging from 650 to 1100°C, in 50°C increments. The camera was positioned at an Object-to-Image Distance (OID) of 140 cm from the blackbody furnace (Fig. 3a). The camera settings were fixed at an ISO of 100, F-numbers of 4.0 and 8.0, and exposure times ranging from 20.0 to 5.0×10^{-4} seconds. Images were captured in RAW format and converted to a linear 16-bit TIFF format using DCraw [22, 23, 25–28], with demosaicing performed using the Variable Number of Gradients (VNG) method.

Multiple images were taken at each blackbody temperature, varying exposure times and F-numbers. Mean pixel intensity values and standard deviations for the red, green, and blue bands in the blackbody furnace region were extracted using OpenCV. The images were filtered to exclude under- or over-saturated pixels, focusing on an intensity range of 0.1% to 90%. An increase

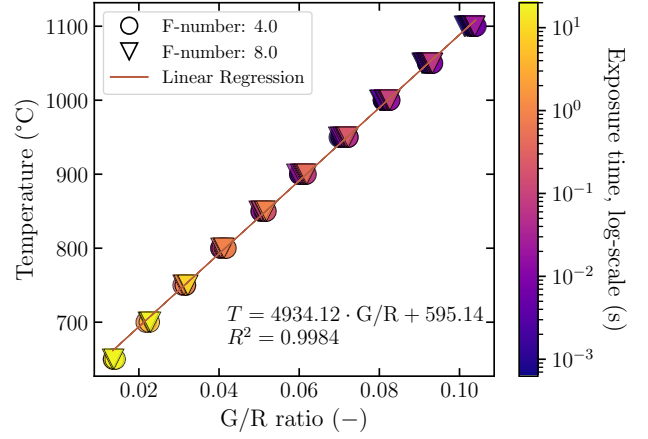


Figure 4: Camera calibration results for color-ratio pyrometry using the ratio of Green and Red color bands, G/R, and the temperature of the blackbody furnace.

in the green/red band intensity ratio (G/R) with blackbody temperature was observed, consistent with previous studies [22–24, 26], while no similar increase was found for any other combination of spectral bands. The G/R ratio was used for camera calibration, revealing a linear correlation with blackbody temperature, as shown in Fig. 4, across F-numbers of 4.0 and 8.0, with ISO = 100. The calibration yielded a high coefficient of determination ($R^2 = 0.9984$) and an RMSE of 5.03°C across the 650–1100°C range. Over 48,000 pixels were analyzed per image, with 129 out of 140 non-saturated images having exposure times below 1.0 second.

For cone heater CRP measurements, the camera was positioned at the same OID as in the blackbody calibration (Fig. 3b). The cone heater was set to temperatures of 750, 800, and 850°C, and multiple images were taken with varying exposure times and F-numbers. The G/R calibration equation from Fig. 4 was then applied to convert the pixel values to surface temperatures of the cone heater, $T_{c,crp}$, resulting in a spatial temperature distribution (thermogram) of the heater surface.

The uncertainty in a CRP measurement depends on the pixel intensities used for the G/R calculation, which are influenced by the camera settings [25]. In this study, all objects analyzed with CRP (i.e., the blackbody furnace and cone heater) are assumed to be in a steady-state condition, allowing the same scene to be imaged under different camera settings. To minimize uncertainty in CRP measurements, images were captured at multiple exposure times to identify the thermogram with the lowest measurement uncertainty. This uncertainty primarily arises from random errors in the green or red pixel intensities (e.g., sensor electrical noise) and factors such as pixel saturation [25]. For blackbody thermograms, the surface temperature is assumed to be isothermal, meaning any spatial variation in pixel values represents measurement error. Consequently, the coefficient of variation for pixel temperatures directly reflects the uncertainty propagated from the variation in green and red pixel intensities. The thermogram with the lowest coefficient of variation in these intensities is thus considered to have the lowest estimated uncertainty. For cone heater thermograms, however, the

heater exhibits a spatial temperature distribution, and other surfaces are visible in the image. These non-heater surfaces are filtered out using a thresholding technique applied to the red and green channels. Due to the temperature variation across the heater, uncertainty in the red and green pixel intensities cannot be assessed in the same manner as for the blackbody. Instead, the thermogram with the lowest coefficient of variation in pixel temperatures is selected, assuming it minimizes measurement error, as observed in the blackbody images.

2.5. Uncertainty Quantification

The incident heat flux to the radiometer was measured directly with the radiometer and theoretically calculated using Eq. (2), which depends on the cone temperature, T_c , and the calculated view factor. The uncertainty in each approach was quantified using the uncertainty propagation law [29], which accounts for the propagation of errors from the measured quantities to the final result. This method applies to both the direct measurement of heat flux and the theoretical calculation, as described below.

2.5.1. Theoretical incident radiant heat flux to the sensor

The uncertainty in the calculated (theoretical) incident heat flux, $\delta \dot{q}_{inc,theo}''$, results from uncertainties in both the temperature spatial variation and the view factor. The propagation of these uncertainties is described by the following equation:

$$\begin{aligned} \frac{\delta \dot{q}_{inc,theo}''}{\dot{q}_{inc,theo}''} &= \sqrt{\left(\frac{\delta F_{g \rightarrow c}}{F_{g \rightarrow c}}\right)^2 + \left(\frac{\partial \dot{q}_{inc,theo}''}{\partial T_c} \frac{\delta T}{\dot{q}_{inc,theo}''}\right)^2} \\ &= \sqrt{\left(\frac{\delta F_{g \rightarrow c}}{F_{g \rightarrow c}}\right)^2 + \left(\frac{4}{T_c} \delta T\right)^2} \end{aligned} \quad (12)$$

Here, $\delta F_{g \rightarrow c}$ and δT represent the uncertainties in the view factor from the heat flux gauge sensor to the cone heater, $F_{g \rightarrow c}$, and in the cone heater temperature, respectively. The values for these uncertainties are provided in Table 1. The uncertainty in the temperature, δT , is determined from the spatial variation of the temperature measured with ratio pyrometry (see Section §3.3). The uncertainty in the view factor, $\delta F_{g \rightarrow c}$, arises from the uncertainty in the position of the sensor, which is controlled by the traverse mechanism. The sensor positions x_g and y_g (cf. Fig. 1) are subject to this uncertainty. The error in the view factor can be estimated using the following expression [29]:

$$\delta F_{g \rightarrow c} = \sqrt{\left(\frac{\partial F_{g \rightarrow c}}{\partial d_1} \delta d_1\right)^2 + \left(\frac{\partial F_{g \rightarrow c}}{\partial x_g} \delta x_g\right)^2 + \left(\frac{\partial F_{g \rightarrow c}}{\partial y_g} \delta y_g\right)^2} \quad (13)$$

This equation accounts for the uncertainties in the sensor's distance from the heater, d_1 , and its positions along the x and y axes.

2.5.2. Experimental incident radiant heat fluxes

The quantification of the uncertainty in the radiometer is the combined standard uncertainty evaluated for uncorrelated input quantities [30, 31]. The uncertainty for the experimental incident heat flux, $\dot{q}_{inc,exp}''$, is calculated using the equation

$\dot{q}_{inc,exp}'' = V/S$, where V is the voltage reading (in volts) from the radiometer and S is its calibration constant (in V/(kW/m²)). The uncertainty in the experimental incident heat flux is estimated by:

$$\delta \dot{q}_{inc,exp}'' = \sqrt{\left(\frac{\partial \dot{q}_{inc,exp}''}{\partial V} \sqrt{\sigma_{V,A}^2 + \sigma_{V,B}^2}\right)^2 + \left(\frac{\partial \dot{q}_{inc,exp}''}{\partial S} \delta_S\right)^2} \quad (14)$$

Here, $\delta \dot{q}_{inc,exp}''$ represents the uncertainty in the experimental incident radiant heat flux, δ_S is the uncertainty in the radiometer's calibration constant, $\sigma_{V,A}$ is the experimental standard deviation and $\sigma_{V,B}$ is the uncertainty of the data acquisition instrumentation (National Instruments), which is negligible and thus neglected for this analysis. The calibration constant uncertainty, δ_S , accounts for the $\pm 3\%$ expanded uncertainty in responsivity specified by the heat flux gauge manufacturer (Medtherm), which is traceable to the National Institute of Standards and Technology (NIST). The experimental standard deviation in the measured voltage arises primarily from random noise but may also include other unresolved effects. One such effect is the presence of small vibrations in the sensor holder, which could introduce uncertainty in its position, leading to fluctuations in the instantaneous heat flux delivered to the sensor and, consequently, variations in the measured voltage. This effect is analogous to the influence of sensor positioning on the uncertainty in the view factor within the model (Eq. 13). While this source of uncertainty exists, it is considered sufficiently small that further investigation is not deemed necessary.

Table 1: Constant uncertainties involved in the uncertainty quantification.

δ_d (mm)	δ_h (mm)	δ_k (mm)	$^*\delta_T$ (°C)	δ_S (V/(kW/m ²))
0.5	1.0	1.0	30 - 33	1.286×10^{-6}

* δ_T represents the CRP standard deviation results range (cf. Section §3.3).

3. Results and discussion

3.1. View factors

In this study, the HC and TC geometries are used to calculate view factors for three configurations: (1) the view factor from a differential element, dA_1 , to the cone heater, $F_{dA_1 \rightarrow c}$, (2) the view factor from a radiometer sensor to the cone, $F_{g \rightarrow c}$, and (3) the view factor from a square sample to the cone heater, $F_{s \rightarrow c}$. These calculations are essential for assessing the variability and uniformity of the incident heat flux distribution and for comparing with previous studies [6–8].

To understand the impact of the cone heater geometry on the spatial uniformity of the incident heat flux, the view factor from a differential element, dA_1 , to the cone heater, $F_{dA_1(x,y) \rightarrow c}$, was calculated over the 100×100 mm² sample area at $d_1 = 25$ mm and $d_1 = 60$ mm, as defined in ASTM E1354 and ISO 5660 standards [3, 4]. The view factors are assumed to exhibit the same relative variation as the incident heat flux, $\dot{q}_{inc,dA_1(x,y)}''$, because the cone heater emissivity, ϵ_c , and temperature, $T_{c,PID}$,

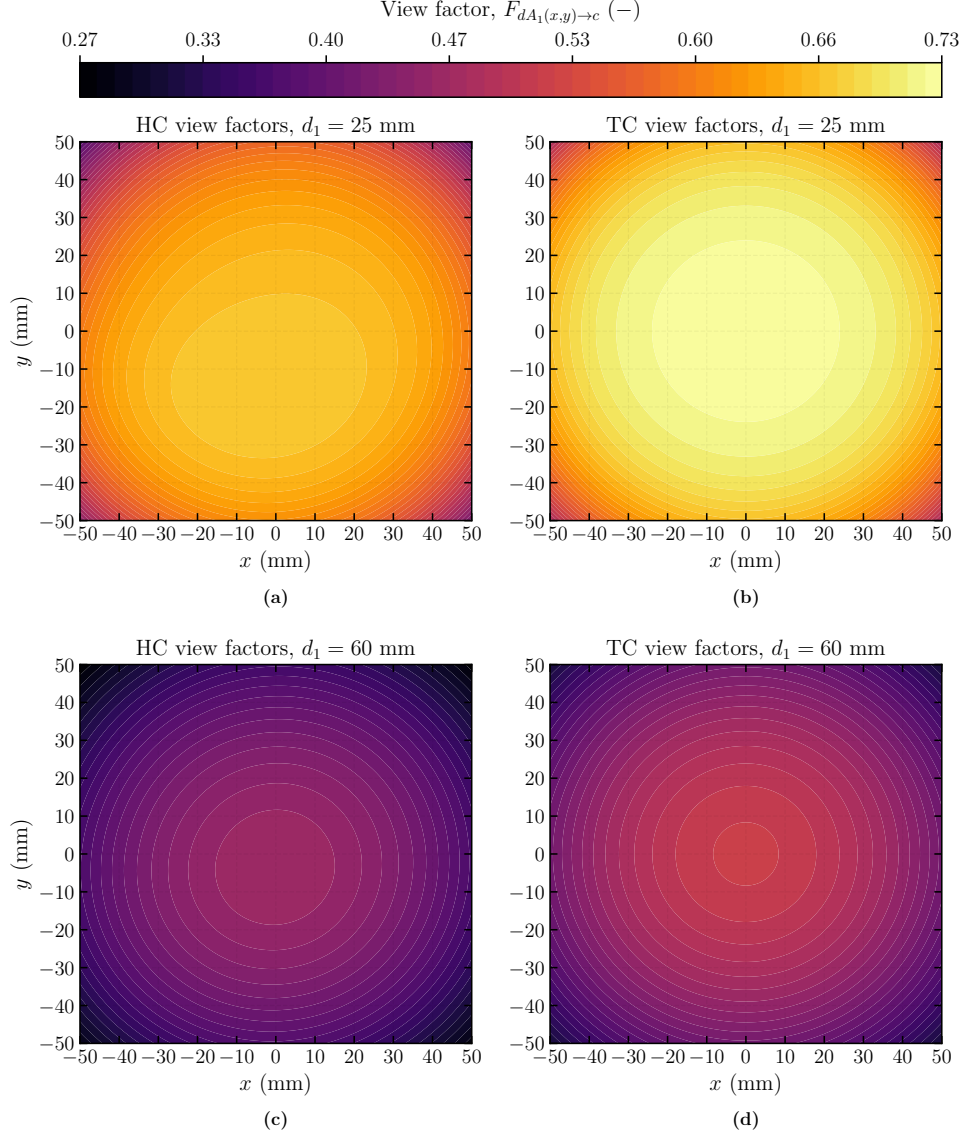


Figure 5: View factor spatial distributions for a $100 \times 100 \text{ mm}^2$ sample at two standard Cone Testing distances for both the helical coils (HC) and truncated cone (TC) configurations.

are considered constants, and the relationship between the incident radiative heat flux to a differential element of the sample is given by the expression $F_{dA_1(x,y) \rightarrow c} = \dot{q}_{inc,dA_1(x,y)}'' / \dot{q}_{emi,c}''$. The view factor spatial distributions, $F_{dA_1(x,y) \rightarrow c}$, are presented in Fig. 5 for both distances, with the left panels showing the HC geometry and the right panels the TC geometry. The distributions were calculated for 5000 points across the sample's surface, with an average point spacing of approximately $dx = dy = 1.41 \text{ mm}$. In this work, the differential element was modeled as a disk with a diameter of $p = 10^{-5} \text{ mm}$.

At $d_1 = 25 \text{ mm}$, Figs. 5a and 5b show that with the TC geometry, the view factor distribution has rotational symmetry, while the HC distribution, although similar, is not symmetric. The peak view factor for HC occurs near $(x, y) \approx (-3.6 \text{ mm}, -16.7 \text{ mm})$ with a value of 0.673, while for TC, the peak is at $(x, y) = (0 \text{ mm}, 0 \text{ mm})$ with a value of 0.726. At $d_1 = 60 \text{ mm}$, the im-

pact of the cone heater coils on the homogeneity of the view factor diminishes. For HC, the highest values are found at $(x, y) \approx (-0.7 \text{ mm}, -3.6 \text{ mm})$ with a value of 0.465. Despite this, the HC configuration still exhibits lower view factor values compared to TC, due to the more realistic representation of the cone heater in the HC model. The $H/19$ (cf. Fig. 1) vertical distance at $r_2(\pi)$ (cf. Eq. 6) causes a lower view factor in the first quadrant (positive x and y) of Figs. 5a and 5c, where the cone heater is slightly higher, contrasting with the TC model's flat heater geometry assumption.

3.1.1. Comparison of view factor calculations and literature values

Previous studies [6, 7] estimated the view factor from a differential element, dA_1 , to a truncated cone-shaped heater. Reference [6] used Stokes' theorem, while Ref. [7] applied an analyt-

ical expression from Ref. [32]. Table 2 compares the view factor values from these studies with results obtained in this work for the position $(x, y, z) = (0, 0, -d_1)$. As previously mentioned, the differential element was modeled as a disk with a diameter of $p = 10^{-5}$ mm, consistent with the approach in Fig. 5. The view factor between the truncated cone (TC) and the disk in this work aligns with literature values at various vertical locations d_1 . However, the HC formulation gives lower values for the same locations. For instance, at $d_1 = 25$ mm, the percentage difference between TC and HC was 8.19%.

Table 2: Comparisons of the view factor from a differential element along the cone centerline ($x = y = 0$) to both the truncated cone heater (TC) and the helical coil heater (HC) at various vertical distances d_1 (cf. Fig. 1). The results from previous works [6, 7] for the TC configuration, along with results from this work for both TC and HC configurations, are shown.

d_1 (mm)	$F_{dA_1 \rightarrow c}$		
	Refs. [6, 7] (TC)	This work (TC)	This work (HC)
10	0.7660	0.7660	0.7136
15	0.7599	0.7599	0.7055
20	0.7461	0.7461	0.6901
25	0.7261	0.7261	0.6690
30	0.7014	0.7014	0.6437

Figure 6a illustrates the view factor from a differential element ($p = 10^{-5}$ mm) to the cone heater as a function of the vertical distance along the centerline, d_1 . In contrast, Fig. 6b presents the view factor from the exposed face of the sample to the cone heater, considering the entire 100×100 mm² sample area (see Appendix B for the view factor formulation). The trends are consistent, with the HC formulation giving lower values, particularly at $d_1 = 25$ mm and $d_1 = 60$ mm, marked by vertical dashed lines in Fig. 6b. These differences arise from the more realistic geometry of the cone heater in the HC model, including the curvature of the coils. For example, the effect of the vertical distance $H/19$ (cf. Fig. 1) at $r_2(\pi)$ (cf. Eq. 6) further influences the HC view factor, resulting in lower values.

At $d_1 = 25$ mm, the view factor from the 100×100 mm² sample to the HC geometry was $F_{s \rightarrow c} = 0.6216$ (cf. Fig. 6b), while the view factor from a differential element at $(0, 0, d_1)$ was $F_{dA_1 \rightarrow c} = 0.6690$ (cf. Table 2). Literature reports $F_{dA_1 \rightarrow c} = 0.78$ [2, 33] using a truncated cone geometry for view factor calculation, showing an 7.13% difference from the TC formulation for a disk of size $p = 12.7$ mm, as used in Ref. [33]. This discrepancy arises from differences in the cone geometry used (e.g., heater radial distances R_1 and R_2 , and vertical distance to the sample, d), as well as the lack of standardized definitions for view factor calculation in this context. It is also notable that comparing the literature's previous result with the proposed HC geometry reveals a 15.30% difference, using the same sample size p .

3.2. Quantifying spatial variation in the view factor distribution

The spatial distribution of the view factor along the x and y axes for both HC and TC formulations is shown in Fig. 7. The

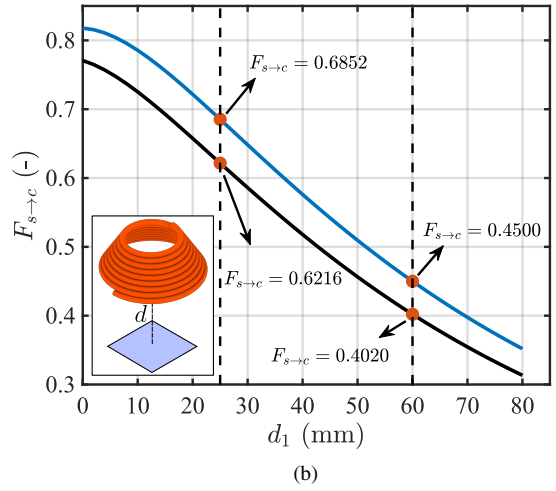
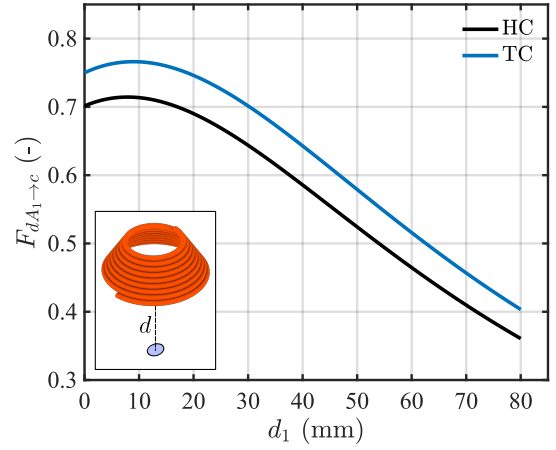


Figure 6: (a) View factor from a differential element ($p = 10^{-5}$ mm) to the cone heater. (b) View factor from the sample to the cone heater. In both cases, the vertical distance along the centerline for the view factor calculation, d , was considered as $d = d_1 + d_2$ (cf. Fig. 1).

TC formulation exhibits equal variations in both directions due to its axial symmetry, while the HC formulation shows greater variation, especially along the y -axis, as seen in the centerline profiles in Fig. 7a. The TC formulation shows minimal variation between $x = y = -20$ mm and $x = y = 20$ mm, whereas significant variations along both axes are evident in Fig. 7b for the HC formulation.

Two metrics were used to quantify the relative variation in the view factor and experimental incident heat flux: P_1 (relative difference) and P_2 (coefficient of variation), defined as:

$$P_1 = \left(\frac{\max(F(x, y)) - \min(F(x, y))}{2\overline{F}(x, y)} \right) \cdot 100\% \quad (15)$$

$$P_2 = \frac{\sigma_{F(x, y)}}{\overline{F}(x, y)} \cdot 100\% \quad (16)$$

Where $F(x, y)$ represents the view factor to the cone heater from a differential element, dA_1 , located at some position (x, y) on a

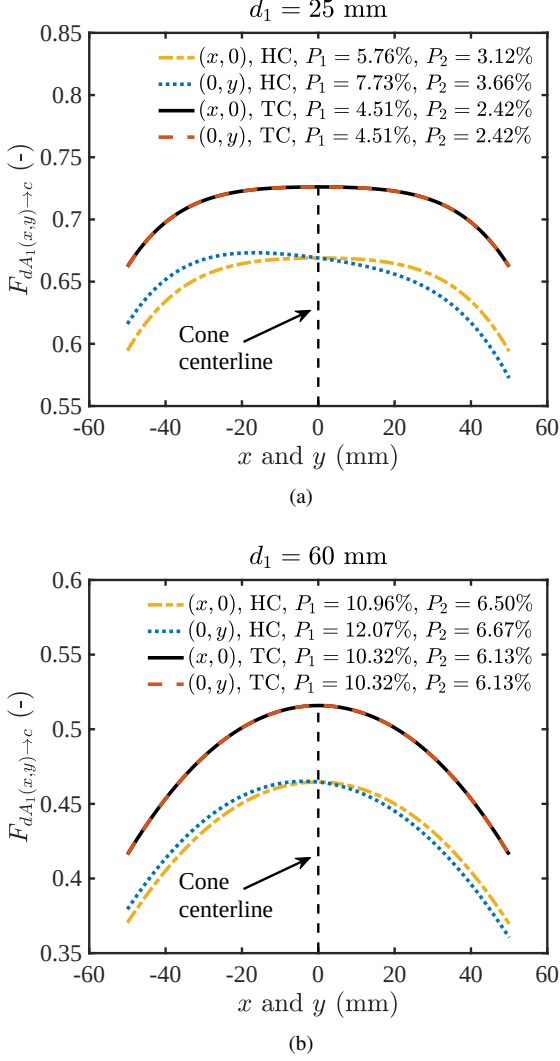


Figure 7: View factor profiles along the sample holder in the x and y axes directions for (a) $d_1 = 25$ mm, and (b) $d_1 = 60$ mm.

defined surface at a constant elevation z (i.e., $F(x, y) = F_{dA_1(x,y) \rightarrow c}$). Additionally, $\sigma_{F(x,y)}$ is the standard deviation of the view factor values, $F(x, y)$, over the specified domain.

The relative variation of P_1 and P_2 was analyzed across the central 50×50 mm² area of the sample surface (cf. Fig. 5), with the results presented in Table 3. For the view factors, both parameters increase as the distance from the cone heater to the target increases. While ASTM E1354 and ISO 5660 standards specify that irradiance variation should not exceed $\pm 2\%$ [2–4], no specific metric is defined for this limit. Table 3 shows that both TC and HC geometries yield P_2 values under 2% at $d_1 = 25$ mm. However, the HC geometry exhibits around 2.5 times the percentage variation compared to the TC geometry. Additionally, P_1 results indicate that only the TC geometry at $d_1 = 25$ mm satisfies both the ASTM E1354 and ISO 5660 limits.

3.3. Heater surface temperature measurements

Color ratio pyrometry (CRP) was applied to images of the vertically positioned cone heater (cf. Fig. 3). Although convective

Table 3: Results of P_1 and P_2 (see Eqs. (15) and (16)) for different geometries and distances for the central area of 50×50 mm² of the contour plots shown in Fig. 5.

Geometry	d_1 (mm)	P_1 (%)	P_2 (%)
HC	25	2.97	1.15
TC	25	1.11	0.44
HC	60	5.51	2.23
TC	60	4.39	1.95

cooling may introduce slight variations, capturing images with the heater in a horizontal configuration was not feasible in the iCone Classic apparatus. However, a separate test on a different cone calorimeter heater confirmed that orientation has a negligible effect on the statistical temperature spatial variation, with mean values differing by less than 1°C. After post-processing the images collected from the main study, the average temperature and standard deviation were extracted from the green and red color channels of the images, and the results are shown in Fig. 8. The first row displays the thermograms obtained from CRP images, and the second row presents the probability density of the pixel temperatures. These results combine CRP statistics with blackbody furnace values at the corresponding set-point temperature. Notably, the average temperature measured by CRP, $\bar{T}_{c,crp}$, is, for all cases, around 50°C higher than the temperature set at the PID controller, $T_{c,PID}$. This discrepancy primarily results from the unknown spectral emissivity of the cone heater in the visible range, leading to a suspected systematic overestimation in CRP measurements. Additionally, the PID controller uses three thermocouples positioned on the backside of the heater, which may not fully capture temperature variations across the entire coil surface. Despite these spatial variations, the thermocouple measurements remain the most accurate for determining the heater temperature, as they directly monitor the coils.

The second row of Fig. 8 compares temperatures obtained from the blackbody calibration source and the cone heater using Kernel Density Estimation (KDE) to illustrate statistical distribution. For instance, when the cone heater was set to 800°C, CRP measured $\bar{T}_{c,crp} = 847.1 \pm 31.1$ °C, while the blackbody furnace gave $\bar{T}_{bb,crp} = 802.5 \pm 10.3$ °C. The blackbody images used in the present analysis (black curves) are different from those used to construct the CRP calibration curve in Fig. 4. These blackbody images, taken at the same set-point temperatures as the cone heater and using camera settings which minimized the measurement error (see Section §2.4.1 for details), are consistent with the method applied to the CRP measurements of the cone heater. Despite using the blackbody for calibration, discrepancies between the CRP and cone heater temperatures suggest that surface conditions, such as emissivity variations, or spectral dependence of the surface properties may affect the CRP measurements. As previously mentioned, the CRP measurements are all roughly 50°C higher due to a suspected bias error introduced by the unknown spectral emissivity in the visible range. While Ref. [9] showed blackbody-like behavior of

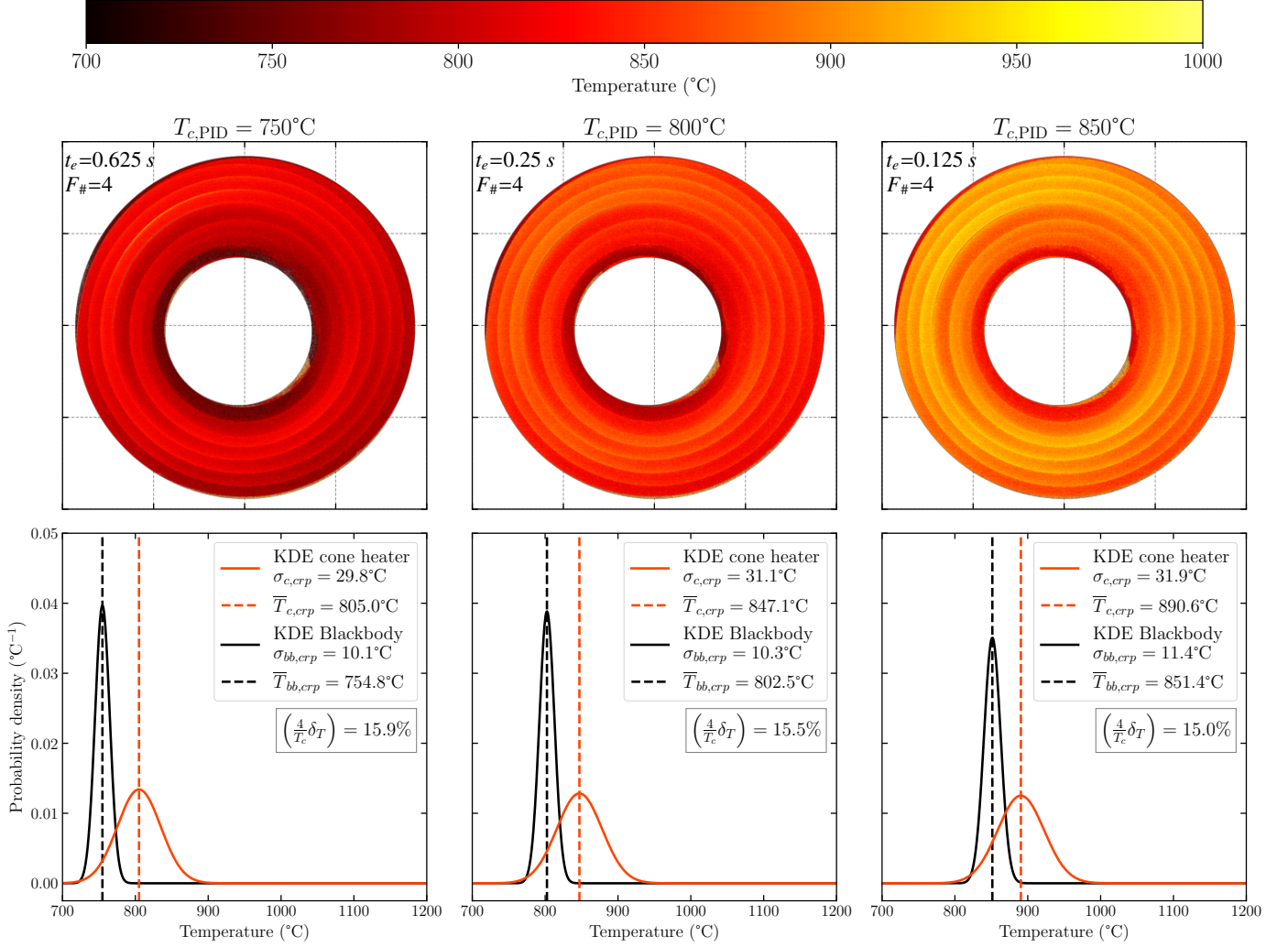


Figure 8: Temperature images of the cone heater, estimated using color ratio pyrometry (CRP), are compared with blackbody furnace results at the same setpoint. It is important to note that the blackbody furnace image results are not those used for the CRP calibration. For all cases, the F-number is 4.0, and the exposure times are 5/8, 1/4, and 1/8 seconds for temperatures of 750, 800, and 850°C, respectively. The value of $4\delta_T/T_c$ represents the contribution of the uncertainty of the cone heater temperature to the relative uncertainty in the theoretical incident heat flux (cf. Section §2.5).

the cone heater in the IR at 858 K, emissivity variations in the visible spectrum could shift measured temperatures.

Given the narrow detection band, this bias is likely small. For some applications 50°C accuracy may be sufficient, but for the goals of this work, this error is too great for the average temperature from CRP to be used. On the other hand, the impact of a possible bias error on the standard deviation of the pixel temperature distribution is expected to be small enough for this value to be used in quantifying the spatial variations in the heater surface temperature. The PID-controlled in-situ measurements, however, remain the most reliable for determining the temperature, as they directly monitor the coils through thermocouples. Therefore, the temperature set at the PID controller is used for calculating the theoretical incident heat flux.

CRP measurements assume uniform emissivity in the green and red color bands and no surface residues affecting pixel intensity. The cone heater coils may not meet these ideal conditions, which could impact measurement accuracy. Nevertheless, CRP

measurements were used to estimate the uncertainty in the emitter temperature, $\sigma_{c,crp} = \delta_T$, which influences the uncertainty in theoretical incident heat flux calculations in Eq. (12).

3.4. Incident heat flux measurements and predictions

Incident heat flux measurements were made along the centerline of the cone heater at eight vertical distances ($d_1 \in \{25, \dots, 60\}$ mm, in 5 mm increments). These measurements were compared to theoretical formulations for both the HC and TC configurations, where a disk with a diameter of $p = 3.175$ mm, matching that of the radiometer sensor used, was employed in the calculation of the view factor, $F_{g \rightarrow c}$. The ratio of experimental to theoretical incident heat flux at these distances along the centerline $(0, 0, z)$ is illustrated in Fig. 9a. Note that, when comparing the theoretical incident heat flux to the experimental data, the vertical distance in the z -direction is expressed as $d = d_1 + d_2 + d_3$ in the view factor model (cf. Fig. 1). The results show improved agreement between exper-

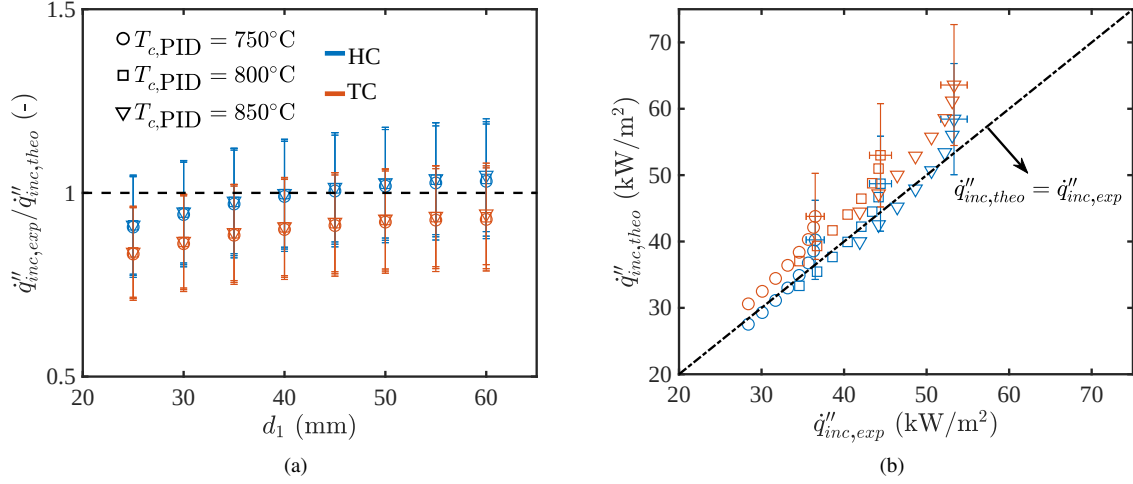


Figure 9: Comparison of experimental and theoretical incident heat fluxes along the cone heater centerline (0, 0, z). Theoretical values were calculated using the cone calorimeter PID controller temperature, $T_{c,PID}$. In (a), the error bars represent the complete theoretical uncertainties, while in (b), only the highest uncertainties are shown, which apply to both the experimental and theoretical data.

imental and theoretical values as d_1 increases, suggesting that sensitivity to the spatial variation of the cone heater temperature diminishes at larger distances, leading to closer convergence (cf. Fig. 9a). On average, the HC configuration demonstrates better agreement, with the ratio of experimental to theoretical heat flux closer to unity across all data compared to the TC configuration. At smaller d_1 , greater deviations are observed, likely due to the amplified effect of geometric details.

Figure 9b compares theoretical incident heat flux (y-axis) to experimental heat flux (x-axis) for both HC and TC configurations. The error bars represent uncertainties, with the largest values highlighted for the shown data. On average, the HC configuration demonstrates better agreement with the experimental data compared to the TC configuration, which shows a slightly higher tendency to overestimate the heat flux. At each cone heater temperature, the largest errors are found at smaller vertical distances d_1 (as shown in Fig. 9a) or higher heat fluxes. At these points, the theoretical incident heat flux also shows the greatest overestimation compared to experimental measurements. While the error bars in Fig. 9 represent combined uncertainties, analysis indicates that the largest contributor to theoretical uncertainty is the temperature spatial variation of the cone heater, particularly the standard deviation of cone heater temperatures obtained via CRP measurements. Experimental uncertainty remains significantly smaller than theoretical uncertainty across all data, with the highest source of uncertainty arising from the calibration of the gauge.

To further explore these effects, the experimental setup was adapted to measure the incident heat flux at various locations within a fixed horizontal plane. For these measurements, the traverse system was calibrated to move in increments of 25 mm within the plane at $z = d_1$. The results for the experimental and theoretical incident radiant heat fluxes are presented in Fig. 10, with the measurement locations, numbered from 1 to 25, and their corresponding (x_g, y_g) coordinates depicted at the top-right

corner of Fig. 10b. Each plot corresponds to a specific cone heater (PID) temperature, $T_{c,PID}$, and vertical distance d_1 , as indicated in the plot titles. Three datasets are shown in each plot: experimental data, and theoretical incident heat fluxes calculated using both the helical coil (HC) and truncated cone (TC) geometries for the view factor. The error bars represent the corresponding maximum uncertainties (cf. Section §2.5). Additionally, each plot shows the root mean square error (RMSE) between the theoretical (HC or TC) and experimental incident heat fluxes.

Notably, the highest uncertainties for all theoretical cases were observed at the center (Location 1), which aligns with the highest theoretical incident heat fluxes occurring at that point. In contrast, the highest experimental uncertainty was found at Location 13, one of the farthest points within the 100×100 mm² area. The uncertainty at this location, compared to points 17, 21, and 25, is likely due to the proximity to the cone heater and the sensitivity of the view factor to positioning in this region (cf. Fig. 5). The results at $d_1 = 25$ mm show that the central 50×50 mm² region (locations 1 to 9) experiences a roughly constant incident heat flux, while outside this area, there is significantly more variation, likely due to the spatial variation of the cone heater temperature and its geometry. However, further away, at $d_1 = 60$ mm, the variation decreases, suggesting that the effect of the cone heater's temperature non-uniformity becomes less significant at greater distances. At $d_1 = 60$ mm, the RMSE values are lower for all cone heater temperatures, indicating that the theoretical models perform better at this distance.

A similar approach to that used for estimating the spatial variation of the view factor in Section §3.2 was employed to quantify P_1 and P_2 for the incident heat flux measurements across the central 50×50 mm² region (Locations 1 to 9), following Eqs. (15) and (16), but substituting $\dot{q}_{inc,exp}''(x, y)$ in place of $F(x, y)$. The values of P_1 and P_2 obtained from these measurements are presented in Table 4. The spatial variation, quantified

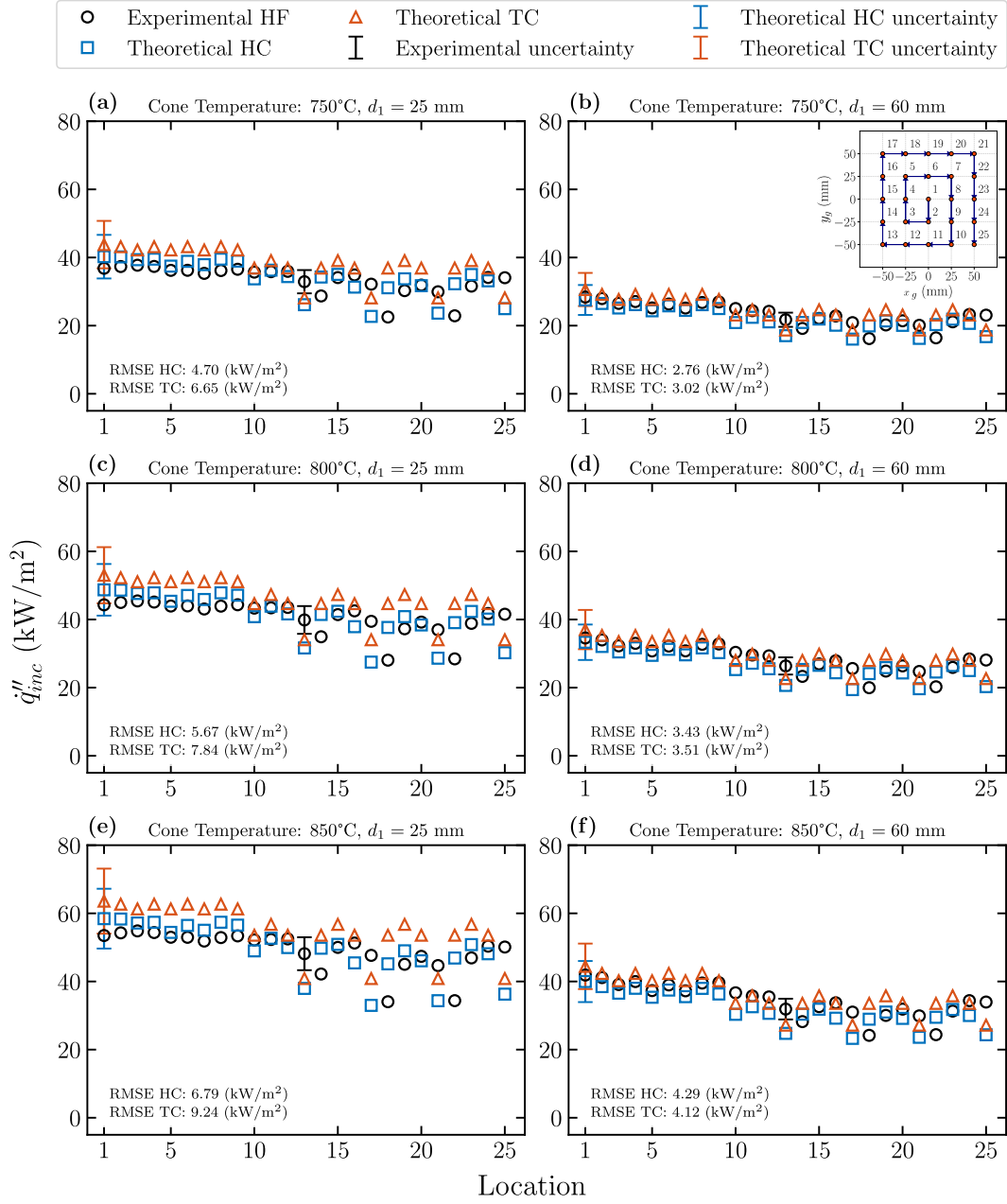


Figure 10: Comparison of experimental and theoretical incident radiant heat fluxes for helical coil (HC) and truncated cone (TC) configurations. The x-axis denotes the measurement locations, as illustrated at the top right of (b).

as P_1 and P_2 , was less pronounced closer to the heater ($d_1 = 25$ mm) compared to the more substantial variation observed at $d_1 = 60$ mm. This trend was consistent across all cone heater temperature set points, $T_{c,PID}$.

An extended analysis of the previous results is presented in Fig. 11, comparing the experimental and theoretical incident heat fluxes. The markers represent different set-point temperatures at the PID controller, with colors indicating the geometry (HC or TC) and closed and open markers denoting distances of $d_1 = 25$ mm and $d_1 = 60$ mm, respectively. Figure 11a focuses on the central 50x50 mm² region, where good agreement is observed, particularly for the theoretical heat flux using the

Table 4: Range of P_1 and P_2 values (see Eqs. (15) and (16)) for incident heat flux measurements across the central 50x50 mm² region (Locations 1 to 9) at distances $d_1 = 25$ mm and $d_1 = 60$ mm. The range represents the minimum and maximum values obtained across all cone heater temperature set points, $T_{c,PID}$.

d_1 (mm)	P_1 (%)	P_2 (%)
25	2.67 – 3.26	1.66 – 2.05
60	5.85 – 6.03	3.88 – 4.05

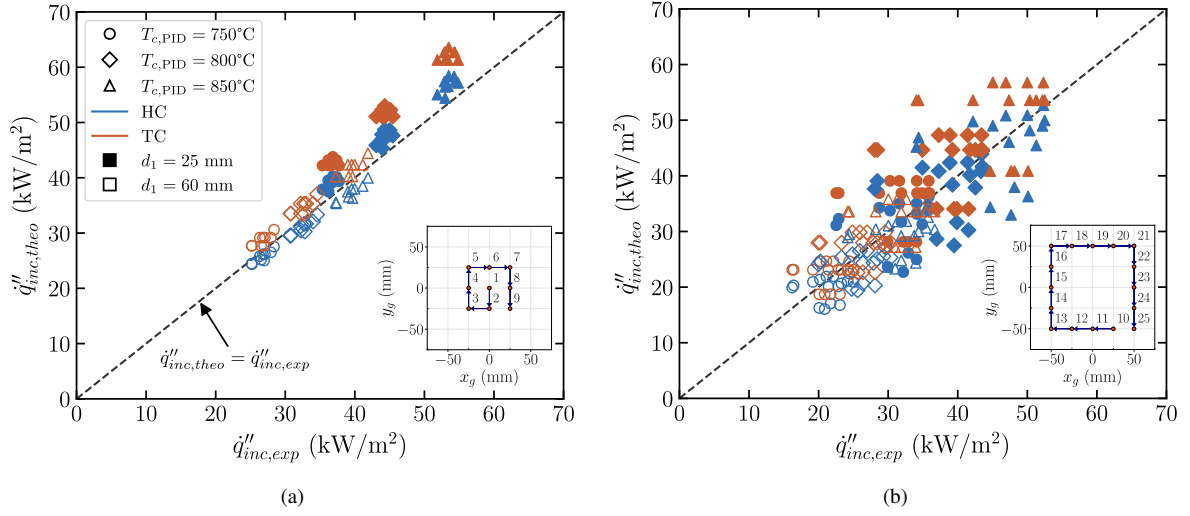


Figure 11: Experimental vs. theoretical incident radiant heat flux comparison. (a) Data for the central $50 \times 50 \text{ mm}^2$ region, and (b) data for the area outside the $50 \times 50 \text{ mm}^2$ region. The markers represent the cone heater temperature ($T_{c,PID}$), with colors indicating the geometry (blue for HC and orange for TC). Closed markers correspond to $d_1 = 25 \text{ mm}$, while open markers correspond to $d_1 = 60 \text{ mm}$.

HC formulation, especially at $d_1 = 60 \text{ mm}$. In contrast, outside the $50 \times 50 \text{ mm}^2$ region (Fig. 11b), the comparison is less consistent, with data becoming more dispersed. This is likely due to the sensitivity of the view factor distribution to location, especially farther from the center or peak value. This can be seen by examining the transitions between different contours in Fig. 5. At the outer corner points, small imprecisions in sensor positioning, including potential minor vibrations, are expected to cause the largest uncertainty in the measured heat flux. For the central region, the HC formulation performed better at all temperatures and vertical distance, which was evaluated with the RMSE.

The results show that the helical coil (HC) model consistently exhibits a lower RMSE compared to the truncated cone (TC) model, confirming that the HC model provides a more accurate representation of the cone calorimeter heater.

4. Conclusions

This work presents an idealized and simplified formulation for the view factor from the cone heater in the cone calorimeter, treating the cone heater as a helical coil (HC). The new formulation was compared with a previous model that approximated the cone heater as a truncated cone (TC). Experimental measurements of incident heat flux validated the HC approach, showing, on average, better agreement with the experimental data compared to the TC formulation. This comparison underscores the effectiveness of the newly proposed HC-based formulation for estimating view factors in the cone calorimeter.

The uncertainty quantification was conducted for both the experimental and theoretical incident radiant heat flux. Experimental uncertainty was primarily influenced by measurement variations and the calibration constant of the heat flux gauge.

For the theoretical uncertainty, the analysis accounted for uncertainties related to the view factor model, including variables such as x_g , y_g , and d_1 , as well as the temperature spatial variation of the coils. The results indicate that the primary source of theoretical uncertainty arises from the spatial variation of the cone heater temperature, characterized using color ratio pyrometry.

These findings suggest that the helical coil geometry provides a more accurate representation of the incident heat flux compared to the truncated cone model. This is especially relevant for improving the precision of heat flux estimates in the cone calorimeter.

Looking ahead, the proposed model could be applied to materials with complex thermal degradation behaviors, such as intumescent coatings and wood, which experience non-uniform heating due to surface expansion or contraction during degradation. In these cases, a larger spacing between the sample and the heater is required. The analytical radiation model developed here offers a reliable methodology for estimating heat flux and assessing its uniformity across the surface, without relying solely on direct experimental measurements. These results contribute to a better understanding of heat flux distribution from the cone calorimeter heater and provide a promising framework for future fire safety assessments and research.

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Supplementary material

All the models used for the view factor calculations in this study are provided in Python and can be accessed in the GitHub repository:

<https://github.com/javaldivia/Cone-Calorimeter-View-Factor>.

The repository includes all relevant scripts, ensuring full transparency and reproducibility of the models employed.

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Appendix A. Truncated cone formulation

Appendix A.1. View factor from the sensor to the cone heater

The gauge’s sensor with center at $(x_g, y_g, -d)$ is represented by:

$$\Gamma_1 : \mathbf{r}_1(u) = \left(\frac{p}{2} \cos u + x_g, \frac{p}{2} \sin u + y_g, -d \right), 0 \leq u \leq 2\pi. \quad (\text{A.1})$$

The top and the bottom contours are given by:

$$\begin{aligned} \Gamma_2 : \mathbf{r}_2(v) &= (R_1 \cos v, R_1 \sin v, H), & 0 \leq v \leq 2\pi \\ \Gamma_3 : \mathbf{r}_3(v) &= ((R_1 + R_2) \cos v, (R_1 + R_2) \sin v, 0), & 2\pi \leq v \leq 0. \end{aligned} \quad (\text{A.2})$$

The distances between the contours (R_{1j}) and the dot product between the tangent vectors ($d\mathbf{r}_1 d\mathbf{r}_j$) are defined as:

$$\begin{aligned} R_{12} &= \|\mathbf{r}_1(u) - \mathbf{r}_2(v)\| \\ R_{13} &= \|\mathbf{r}_1(u) - \mathbf{r}_3(v)\| \\ d\mathbf{r}_1 d\mathbf{r}_2 &= \frac{p}{2} R_1 \cos(u-v) du dv \\ d\mathbf{r}_1 d\mathbf{r}_3 &= \frac{p}{2} (R_1 + R_2) \cos(u-v) du dv \end{aligned} \quad (\text{A.3})$$

Finally, the expression that allows to estimate the view factor between the truncated cone and a disk (sensor) is given by:

$$F_{g \rightarrow c} = \frac{2}{\pi^2 p^2} \left(\oint_0^{2\pi} \oint_0^{2\pi} \ln(R_{12}) d\mathbf{r}_1 d\mathbf{r}_2 + \oint_{2\pi}^0 \oint_0^{2\pi} \ln(R_{13}) d\mathbf{r}_1 d\mathbf{r}_3 \right) \quad (\text{A.4})$$

Appendix A.2. View factor from the sample to the cone heater

The sample is given by a square plate of side length L which is represented by its contours as:

$$\begin{aligned} C_1 : \mathbf{r}_1(u) &= \left(\frac{L}{2} - Lu, \frac{L}{2}, -d \right), \quad 0 \leq u \leq 1. \\ C_2 : \mathbf{r}_2(u) &= \left(-\frac{L}{2}, \frac{L}{2} - Lu, -d \right), \quad 0 \leq u \leq 1. \\ C_3 : \mathbf{r}_3(u) &= \left(-\frac{L}{2} + Lu, -\frac{L}{2}, -d \right), \quad 0 \leq u \leq 1. \\ C_4 : \mathbf{r}_4(u) &= \left(\frac{L}{2}, -\frac{L}{2} + Lu, -d \right), \quad 0 \leq u \leq 1. \end{aligned} \quad (\text{A.5})$$

The contours of the truncated cone for the top and bottom are given by:

$$\begin{aligned} \Gamma_5 : \mathbf{r}_5(v) &= (R_1 \cos v, R_1 \sin v, H), \quad 0 \leq v \leq 2\pi. \\ \Gamma_6 : \mathbf{r}_6(v) &= ((R_1 + R_2) \cos v, (R_1 + R_2) \sin v, 0), \quad 2\pi \leq v \leq 0, \end{aligned} \quad (\text{A.6})$$

The distances between the contours (R_{ij}) and the dot product between the tangent vectors ($d\mathbf{r}_i d\mathbf{r}_j$) are defined as:

$$\begin{aligned} R_{ij} &= \|\mathbf{r}_i(u) - \mathbf{r}_j(v)\|, \quad i = 1, 2, 3, 4, j = 5, 6. \\ d\mathbf{r}_i d\mathbf{r}_j &= \mathbf{r}'_i(u) \cdot \mathbf{r}'_j(v) du dv, \quad i = 1, 2, 3, 4, j = 5, 6. \end{aligned} \quad (\text{A.7})$$

Finally,

$$\begin{aligned} F_{s \rightarrow c} &= \frac{1}{2\pi L^2} \left(\oint_0^{2\pi} \int_0^1 \ln(R_{15}) d\mathbf{r}_1 d\mathbf{r}_5 + \oint_{2\pi}^0 \int_0^1 \ln(R_{16}) d\mathbf{r}_1 d\mathbf{r}_6 \right. \\ &\quad + \oint_0^{2\pi} \int_0^1 \ln(R_{25}) d\mathbf{r}_2 d\mathbf{r}_5 + \oint_{2\pi}^0 \int_0^1 \ln(R_{26}) d\mathbf{r}_2 d\mathbf{r}_6 \\ &\quad + \oint_0^{2\pi} \int_0^1 \ln(R_{35}) d\mathbf{r}_3 d\mathbf{r}_5 + \oint_{2\pi}^0 \int_0^1 \ln(R_{36}) d\mathbf{r}_3 d\mathbf{r}_6 \\ &\quad \left. + \oint_0^{2\pi} \int_0^1 \ln(R_{45}) d\mathbf{r}_4 d\mathbf{r}_5 + \oint_{2\pi}^0 \int_0^1 \ln(R_{46}) d\mathbf{r}_4 d\mathbf{r}_6 \right) \end{aligned} \quad (\text{A.8})$$

Appendix B. Helical coil formulation

Appendix B.1. View factor from the sample to the cone heater

The contours of the square sample with side length L are defined as:

$$\begin{aligned} C_1 : \mathbf{r}_1(u) &= \left(\frac{L}{2} - Lu, \frac{L}{2}, -d \right), \quad 0 \leq u \leq 1. \\ C_2 : \mathbf{r}_2(u) &= \left(-\frac{L}{2}, \frac{L}{2} - Lu, -d \right), \quad 0 \leq u \leq 1. \\ C_3 : \mathbf{r}_3(u) &= \left(-\frac{L}{2} + Lu, -\frac{L}{2}, -d \right), \quad 0 \leq u \leq 1. \\ C_4 : \mathbf{r}_4(u) &= \left(\frac{L}{2}, -\frac{L}{2} + Lu, -d \right), \quad 0 \leq u \leq 1. \end{aligned} \quad (\text{B.1})$$

Similarly, the cone heater contours are given by:

$$\begin{aligned} \Gamma_5 : \mathbf{r}_5(v) &= \left(f(v) \cos v, f(v) \sin v, \frac{Hv}{19\pi} \right), \quad 17\pi \leq v \leq 0 \\ \Gamma_6 : \mathbf{r}_6(v) &= \mathbf{r}_5(v), \quad 2\pi \leq v \leq 19\pi \\ \Gamma_7 : \mathbf{r}_7(v) &= \left(\frac{D_{elem}}{2} \cos v - R_1 - \frac{18R_2}{19}, 0, \frac{D_{elem}}{2} \sin v + \frac{H}{19} \right) \\ \Gamma_8 : \mathbf{r}_8(v) &= \left(\frac{D_{elem}}{2} \cos v + R_1 + \frac{R_2}{19}, 0, \frac{D_{elem}}{2} \sin v + \frac{18H}{19} \right) \end{aligned} \quad (\text{B.2})$$

where the linear function, $f(v)$, is given by:

$$f(v) = -R_1 - R_2 + \frac{R_2 v}{19\pi} \quad (\text{B.3})$$

The domain of the contours Γ_7 and Γ_8 are $\alpha - \pi \leq v \leq \alpha$ and $\pi - \alpha \leq v \leq 2\pi - \alpha$, respectively, where $\alpha = \arctan(H/R_2)$. The distances between the contours (R_{ij}) and the dot product between the tangent vectors ($d\mathbf{r}_i d\mathbf{r}_j$) are defined as:

$$\begin{aligned} R_{ij} &= \|\mathbf{r}_i(u) - \mathbf{r}_j(v)\|, \quad i = 1, 2, 3, 4, j = 5, 6, 7, 8. \\ d\mathbf{r}_i d\mathbf{r}_j &= \mathbf{r}'_i(u) \cdot \mathbf{r}'_j(v) du dv, \quad i = 1, 2, 3, 4, j = 5, 6, 7, 8. \end{aligned} \quad (\text{B.4})$$

Finally,

$$\begin{aligned} F_{s \rightarrow c} &= \frac{1}{2\pi L^2} \left(\int_{17\pi}^0 \int_0^1 \ln(R_{15}) d\mathbf{r}_1 d\mathbf{r}_5 + \int_{2\pi}^{19\pi} \int_0^1 \ln(R_{16}) d\mathbf{r}_1 d\mathbf{r}_6 \right. \\ &\quad + \int_{\alpha-\pi}^{\alpha} \int_0^1 \ln(R_{17}) d\mathbf{r}_1 d\mathbf{r}_7 + \int_{\pi-\alpha}^{2\pi-\alpha} \int_0^1 \ln(R_{18}) d\mathbf{r}_1 d\mathbf{r}_8 \\ &\quad + \int_{17\pi}^0 \int_0^1 \ln(R_{25}) d\mathbf{r}_2 d\mathbf{r}_5 + \int_{2\pi}^{19\pi} \int_0^1 \ln(R_{26}) d\mathbf{r}_2 d\mathbf{r}_6 \\ &\quad + \int_{17\pi}^0 \int_0^1 \ln(R_{35}) d\mathbf{r}_3 d\mathbf{r}_5 + \int_{2\pi}^{19\pi} \int_0^1 \ln(R_{36}) d\mathbf{r}_3 d\mathbf{r}_6 \\ &\quad + \int_{\alpha-\pi}^{\alpha} \int_0^1 \ln(R_{37}) d\mathbf{r}_3 d\mathbf{r}_7 + \int_{\pi-\alpha}^{2\pi-\alpha} \int_0^1 \ln(R_{38}) d\mathbf{r}_3 d\mathbf{r}_8 \\ &\quad \left. + \int_{17\pi}^0 \int_0^1 \ln(R_{45}) d\mathbf{r}_4 d\mathbf{r}_5 + \int_{2\pi}^{19\pi} \int_0^1 \ln(R_{46}) d\mathbf{r}_4 d\mathbf{r}_6 \right). \end{aligned} \quad (\text{B.5})$$